MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

* NETWORK, RESISTOR, POWER, KEY CODE RUM

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the procurement of an electronic resistor network module.

1.2 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Name	Network, resistor
Family	Power
Key code	RUM
Size	1A
Weight	35 grams maximum
* Failure rate	0.080 failures/10^6 hours maximum
Environment	Class I as specified in MIL-STD-1389
Maximum power dissipation	2.5 W
Circuits per module	13
Resistors per circuit	Two 20 kΩ, 1/10 W

* 1.3 Module function. This module contains twenty-six 20 kΩ ±1.75%, 1/10 W resistors. These resistors are connected to make available 13 sets, of two resistor each, for use as input summing resistors.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command, ATTN: ELEX 5043, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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* 2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.
MIL-STD-1665 - Test equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, drawings, and publication required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

* 3.3 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 0.080 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

* 3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be RUM.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/204-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

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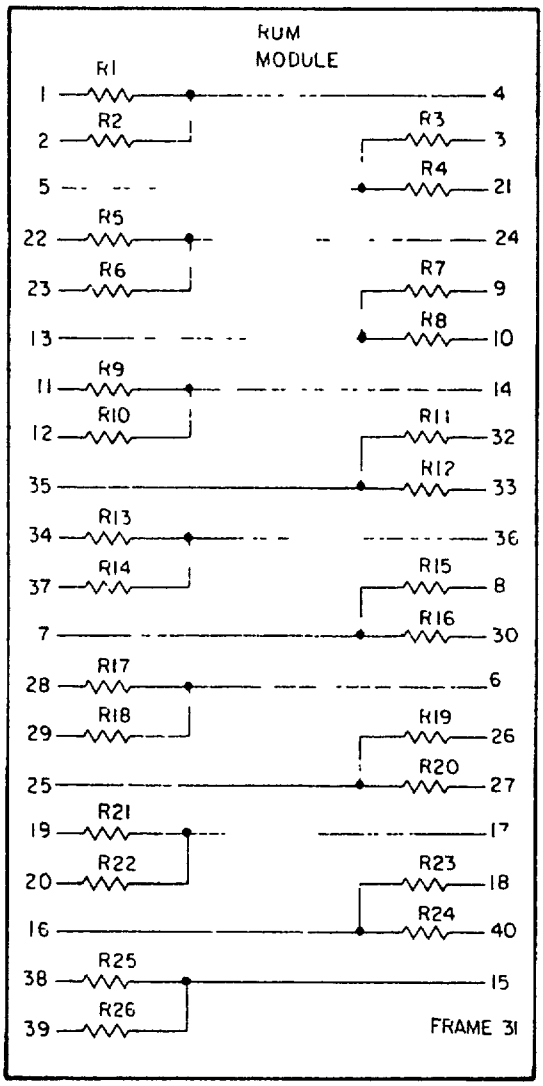
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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-2/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A13	Isolation	II	X	10.0	∞	1.0	∞	10.0	∞	1.0	∞	MΩ
B	Continuity	III	S	0	1.0	0	1.0	0	1.0	0	1.0	Ω
C1-C26	Resistance	IV	X	19.80	20.20	19.70	20.30	19.75	20.25	19.65	20.35	kΩ
D1-D13	Network crosstalk	V	S	0	5.60	0	5.60	0	5.60	0	5.60	mVrms

1/ "X" parameters shall be 100 percent tested during module acceptance test.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

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* NOTE: R1-R26 = 20.0 kΩ, 1/10 W.

FIGURE 1. RUM module function diagram, resistor network.

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3.7 Acceptance test. All production modules shall be tested in accordance with notes 1 and 2 of table I.

* 3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. The following test equipment is necessary to conduct acceptance tests for this module. Test equipment shall be as specified in MIL-STD-1665 and that listed below:

Equipment	MIL-STD-1665 designator	No. required (maximum per test figure)
DC power supply	Item P01	1
Megohm bridge	Item R06	1
Ohmmeter	Item R01	1
Ohmmeter	Item R04	1
Sine wave generator	Item G02	1
Wave analyzer	Item B02	1

4.4 Test procedures. The module shall be tested as specified in table I.

* 4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity (SEMP-QAA) and they conform to the requirements of MIL-M-28787.

* 4.4.2 Order of testing. Test conditions A through D shall be performed in any order with the following exceptions. If test condition A is part of a sequence of tests, the order of testing shall be as follows A where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through D in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test condition C shall be performed using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 2. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

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* 4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests B and C shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests B through D shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

* 5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Part number M28787/204-1 supersedes part number M28787/204. The two parts are interchangeable.

6.4 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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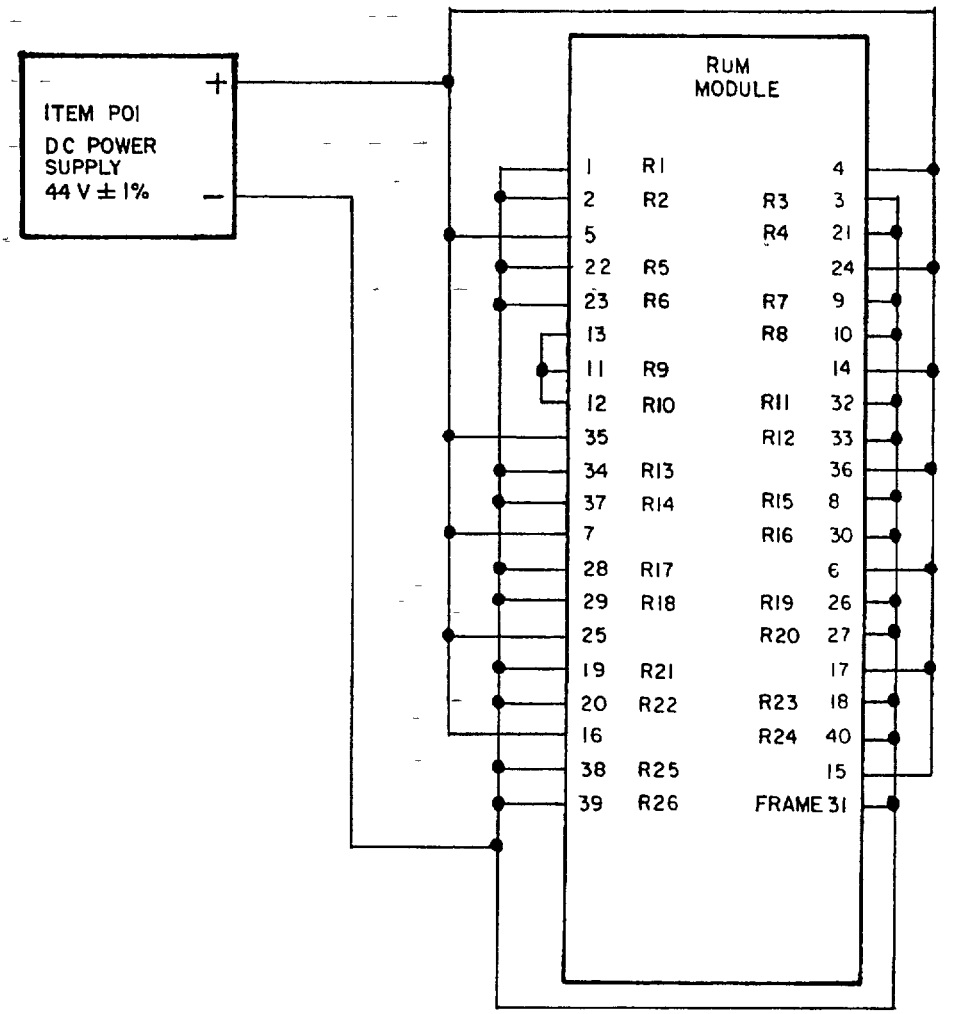


FIGURE 2 Operating temperature, component temperature, vibration, and life test circuit.

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TABLE II. Isolation test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Pin 31 to all others	1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,32,33,34,35,36,37,38,39,40	31	10.0	∞	MΩ	3
A2	Circuit to circuit	3,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,32,33,34,35,36,37,38,39,40	1,2,4				
A3		6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,22,23,24,25,26,27,28,29,30,32,33,34,35,36,37,38,39,40	3,5,21				
A4	Circuit to circuit	6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,25,26,27,28,29,30,32,33,34,35,36,37,38,39,40	22,23,24	10.0	∞	MΩ	3

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* TABLE II. Isolation test - Continued.

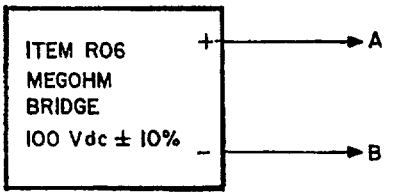
Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A5	Circuit to circuit	6,7,8,11, 12,14,15, 16,17,18, 19,20,25, 26,27,28, 29,30,32, 33,34,35, 36,37,38, 39,40	9,10,13	10.0	∞	MΩ	3
A6		6,7,8,15, 16,17,18, 19,20,25, 26,27,28, 29,30,32, 33,34,35, 36,37,38, 39,40	11,12,14				
A7		6,15,16,17, 18,19,20, 25,26,27, 28,29,32, 33,34,35, 36,37,38, 39,40	7,8,30				
A8	Circuit to circuit	15,16,17, 18,19,20, 25,26,27, 32,33,34, 35,36,37, 38,39,40	6,28,29	10.0	∞	MΩ	3

*

TABLE II. Insolation test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A9	Circuit to circuit	15,16,17, 18,19,20, 32,33,34, 35,36,37, 38,39,40	25,26,27	10.0	∞	MΩ	3
A10		15,16,18, 32,33,34, 35,36,37, 38,39,40	17,19,20				
A11		15,32,33, 34,35,36, 37,38,39	16,18,40				
A12		32,33,34, 35,36,37	15,38,39				
A13	Circuit to circuit	32,33,35	34,36,37	10.0	∞	MΩ	3

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WARNING:

When using the megohm bridge, high voltage may be present at any of the output binding posts. Always set the function switch of the megohm bridge to discharge before connecting or disconnecting the test leads

RUM MODULE			
1	R1		4
2	R2	R3	3
5		R4	21
22	R5		24
23	R6	R7	9
13		R8	10
11	R9		14
12	R10	R11	32
35		R12	33
34	R13		36
37	R14	R15	8
7		R16	30
28	R17		6
29	R18	R19	26
25		R20	27
19	R21		17
20	R22	R23	18
16		R24	40
38	R25		15
39	R26	FRAME	31

*FIGURE 3 Isolation test circuit for table II.

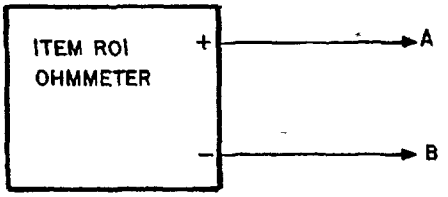
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TABLE III. Continuity test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B	Continuity	31	Frame	0.0	1.0	Ω	4

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NOTE Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

RUM MODULE			
1	R1		4
2	R2	R3	3
5		R4	21
22	R5		24
23	R6	R7	9
13		R8	10
11	R9		14
12	R10	R11	32
35		R12	33
34	R13		36
37	R14	R15	8
7		R16	30
28	R17		6
29	R18	R19	26
25		R20	27
19	R21		17
20	R22	R23	18
16		R24	40
38	R25		15
39	R26	FRAME	31

*FIGURE 4 Continuity test circuit for table III.

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TABLE IV. Resistance test.

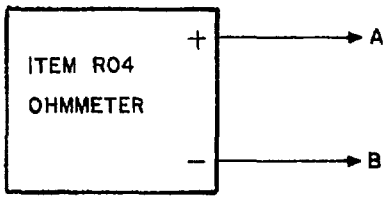
Test condition	Parameter	Connect		Initial limits 25° +5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	Resistance	1	4	19.80	20.20	kΩ	5
C2		2	4				
C3		3	5				
C4		21	5				
C5		22	24				
C6		23	24				
C7		9	13				
C8		10	13				
C9		11	14				
C10		12	14				
C11		32	35				
C12		33	35				
C13		34	36				
C14		37	36				
C15		8	7				
C16		30	7				
C17		28	6				
C18		29	6				
C19	Resistance	26	25	19.80	20.20	kΩ	5

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TABLE IV. Resistance test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C20	Resistance	27	25	19.80	20.20	kΩ	5
C21		19	17				
C22		20	17				
C23		18	16				
C24		40	16				
C25		38	15				
C26	Resistance	39	15	19.80	20.20	kΩ	5

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RUM MODULE

1	R1		4
2	R2	R3	3
5		R4	21
22	R5		24
23	R6	R7	9
13		R8	10
11	R9		14
12	R10	R11	32
35		R12	33
34	R13		36
37	R14	R15	8
7		R16	30
28	R17		6
29	R18	R19	26
25		R20	27
19	R21		17
20	R22	R23	18
16		R24	40
38	R25		15
39	R26	FRAME	31

FIGURE 5 Resistance test circuit for table IV

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TABLE V. Network crosstalk test.

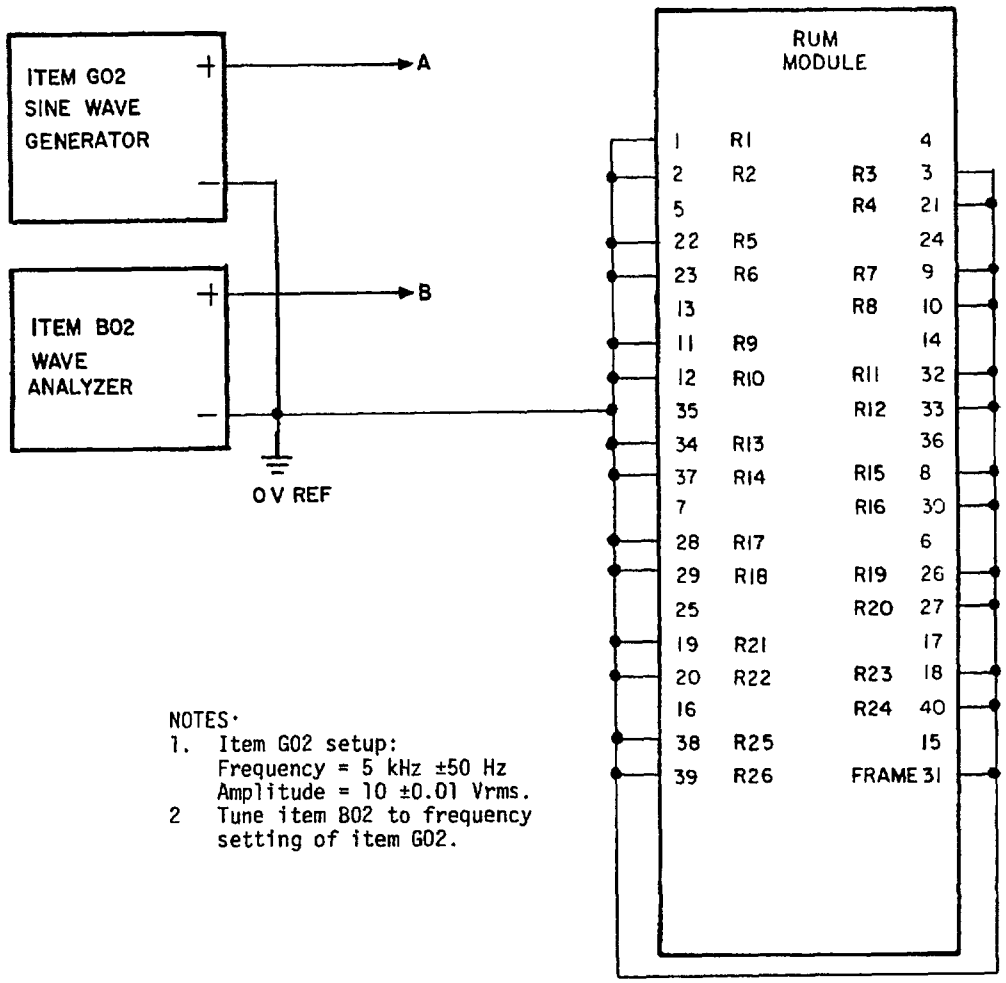
Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D1	Network crosstalk	5,6,7,13, 14,15,16, 17,24,25, 35,36	4	0	5.60	mVrms	6
D2		4,6,7,13, 14,15,16, 17,24,25, 35,36	5				
D3		4,5,7,13, 14,15,16, 17,24,25, 35,36	6				
D4		4,5,6,13, 14,15,16, 17,24,25, 35,36	7				
D5		4,5,6,7, 14,15,16, 17,24,25, 35,36	13				
D6		4,5,6,7, 13,15,16, 17,24,25, 35,36	14				
D7	Network crosstalk	4,5,6,7, 13,14,16, 17,24,25, 35,36	15	0	5.60	mVrms	6

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* TABLE V. Network crosstalk test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D8	Network crosstalk	4,5,6,7, 13,14,15, 17,24,25, 35,36	16	0	5.60	mVrms	6
D9		4,5,6,7, 13,14,15, 16,24,25, 35,36	17				
D10		4,5,6,7, 13,14,15, 16,17,25, 35,36	24				
D11		4,5,6,7, 13,14,15, 16,17,24, 35,36	25				
D12		4,5,6,7, 13,14,15, 16,17,24, 25,36	35				
D13	Network crosstalk	4,5,6,7, 13,14,15, 16,17,24, 25,35	36	0	5.60	mVrms	6

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NOTES:
1. Item G02 setup:
   Frequency = 5 kHz ±50 Hz
   Amplitude = 10 ±0.01 Vrms.
2 Tune item B02 to frequency setting of item G02.

*FIGURE 6 Network crosstalk circuit for table V.

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